AD 502 – Machine Learning

Pre-Reading Material – Lecture 3
Supervised & Unsupervised Machine Learning

Estimated Reading Time: 20–25 minutes
Reference Reading: IIT Madras NPTEL Lectures by Prof. Balaraman Ravindran; "What is supervised learning?"

Why This Topic Matters

Supervised and unsupervised learning are the two most fundamental paradigms in machine learning. Whether a system is predicting stock prices, filtering spam emails, or discovering hidden customer segments, it relies on one of these approaches. Understanding when to use **labeled data** and when to discover patterns from **unlabeled data** is a fundamental skill for every machine learning practitioner.

Learning Outcomes

After completing this pre-reading, you should be able to:

- **Differentiate** between supervised and unsupervised learning based on the type of data used.
- **Distinguish** between classification and regression in supervised learning.
- **Identify** key unsupervised learning tasks, including clustering and association rule mining.
- **Explain** the concepts of overfitting and inductive bias.

Activate Your Prior Knowledge

1. **How do streaming services recommend "More Like This"?**

   Think about how movies are grouped together. Are they grouped because someone has labeled them (supervised learning), or does the system discover similar viewing patterns among millions of users (unsupervised learning)?

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2. **How does a bank detect a suspicious transaction?**

   Banks analyze millions of previous transactions that have already been identified as either **valid** or **fraudulent**. Consider how this historical information helps classify a new, unseen transaction.

3. **Can a computer discover groups you did not know existed?**

   Imagine you have data for 100,000 retail customers without any predefined customer categories. Reflect on how an algorithm might group them based solely on their purchasing behavior.

Prerequisite Concepts

You are expected to be familiar with representing data as **points in a coordinate space** (for example, using age and income as the X and Y axes). You should also understand basic iterative processes and the concept of **prediction error**, which is the difference between a predicted value and the actual value.

Core Concepts

1. Supervised Learning: Learning with Labels

In supervised learning, the algorithm is provided with **labeled training data**, where each input (**X**) is paired with its corresponding target output (**Y**).

- **Classification:** The output is a **discrete or categorical label** (e.g., *Will Buy* or *Will Not Buy*).
- **Regression:** The output is a **continuous numerical value** (e.g., predicting house prices or temperature).
- **Learning Process:** The model makes a prediction (**Ŷ**), compares it with the actual target (**Y**), computes the **prediction error**, and updates its parameters to improve future predictions.

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2. Unsupervised Learning: Discovering Patterns

Unsupervised learning works with **unlabeled data**. Instead of predicting target values, the objective is to discover hidden structures or meaningful patterns within the data.

- **Clustering:** Groups similar data points together while also identifying **outliers**, which are observations that do not belong to any cluster.
- **Association Rule Mining:** Discovers frequent patterns and **conditional relationships** among items (e.g., *Customers who buy bread are also likely to buy milk*).

3. Generalization, Bias, and Overfitting

- **Inductive Bias:** The set of assumptions a learning algorithm makes (for example, assuming that a relationship is approximately linear) to generalize from limited training data to unseen examples.
- **Overfitting:** A situation in which a model becomes overly complex and learns the **noise** in the training data instead of the underlying pattern, resulting in poor performance on new, unseen data.

Guided Reading Questions

1. What is the defining difference between **classification** and **regression**?
2. Why is a **test set** or **validation set** used in supervised learning?
3. In clustering, what are **outliers**, and why might they be important?
4. What does **Market Basket Analysis** refer to in the context of association rule mining?
5. How can **linear regression** be used to solve a classification problem?

Reflection Activity

Write a **150–200 word reflection** on a business problem in an industry of your choice (e.g., healthcare, e-commerce, or manufacturing). Describe one way you could use

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supervised learning to solve a specific task and one way you could use **unsupervised learning** to gain a different insight from the same data source.

Self-Assessment

Answer the following before class:

1. Supervised learning requires a training set consisting of __________ and __________ pairs.
2. If you are predicting the exact temperature (in °C) of a room, you are performing a __________ task.
3. The process of grouping image pixels to discover regions such as *sky* or *sand* is called __________.
4. __________ occurs when an algorithm memorizes the training data, including its noise.
5. Is **Association Rule Mining** a supervised or an unsupervised learning task? __________

5. Why is there no single best Machine Learning algorithm?

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